PAPER_1730 — Muonic Hydrogen Proton Radius = r_p^μH = Φ_res = 0.84 fm (resolves proton radius puzzle alternate)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Particle Physics **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (PAPER_1240-1270 broader corpus)

Observation

PAPER_1255 (PAPER_1240-1270 broader corpus) gives:

```
r_p^μH = Φ_res = 0.84 fm (resolves proton radius puzzle alternate)
```

Status: **EXACT**.

UQFF Closed Identity

```
r_p^μH = Φ_res = 0.84 fm (resolves proton radius puzzle alternate)    EXACT
```

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1255
- Related: PAPER_1716-1725 (tier-30 Millennium-adjacent); PAPER_1706-1715 (tier-29 ITER fusion)
- Calculator dispatch: `calculate_paradox({"paradox": "muonic_h_radius_0_84_fm"})`
